Yijun YUAN

941201yuan@gmail.com
https://yijunyuan.github.io/
0000-0001-6571-6980



Research Interests

- **p-adic Galois representations**
- **p-adic transcendental number theory**
- **Formalization of arithmetic geometry**

Employment

2025–2027 **Postdoctoral Fellow, Institute for Theoretical Sciences, Westlake University**
Advisor: Huayi Chen

Professional Position

2026– **Member, Committee on Mathematics and Artificial Intelligence, Chinese Mathematical Society**

Education

2022–2025 **Ph.D., Yau Mathematical Sciences Center, Tsinghua University**
Pure Mathematics
Thesis title: *The Field of Norms and $(\varphi, \Gamma_{\mathfrak{F},K})$-Modules for p-Adic False Tate Curve Extensions.*
Advisor: Lei Fu / Haoran Wang

2018 – 2021 **M.Sc., Shanghai Center for Mathematical Sciences, Fudan University**
Pure Mathematics
Thesis title: *Uniformizer of the False Tate Curve Extension of $\mathbb{Q}_p$.*
Advisor: Shanwen Wang / Quanshui Wu

2014 – 2018 **B.S., School of Mathematical Sciences, East China Normal University**
Mathematics and Applied Mathematics

Research Publications

Journal Articles

1. **Y. Yuan**, "Formalization of non-archimedean functional analysis 1: Spherically complete spaces," *Journal of Symbolic Logic*, 2026. DOI: 10.1017/jsl.2026.10251
2. S. Wang and **Y. Yuan**, "Hyper-algebraic invariants of p-adic algebraic numbers," *Canadian Mathematical Bulletin*, 2026. DOI: 10.4153/S0008439526102252
3. S. Wang and **Y. Yuan**, "Truncated expansion of ζ_{p^n} in the p-adic mal'cev-neumann field," *Chinese Annals of Mathematics, Series B*, 2026, To appear.

4. S. Wang and **Y. Yuan**, "Uniformizer of the false Tate curve extension of $\mathbb{Q}_p$ (II)," *International Journal of Number Theory*, vol. 20, no. 01, pp. 1–18, 2024. DOI: 10.1142/S1793042124500015
5. S. Wang and **Y. Yuan**, "Uniformizer of the false Tate curve extension of $\mathbb{Q}_p$," *The Ramanujan Journal*, vol. 58, no. 2, pp. 549–595, 2022. DOI: 10.1007/s11139-021-00473-0

Preprints (all submitted)

1. **Y. Yuan**, *Non-liftability of the field of norms of the p-adic false tate curve extension*, 2026. arXiv: 2610.35006 [math.NT].
2. S. Wang and **Y. Yuan**, *The order types of the supports of p-adic algebraic hahn series*, 2026. arXiv: 2610.13014 [math.NT].
3. S. Wang and **Y. Yuan**, *p-adic hahn series with sparse support*, 2026. arXiv: 2607.06944 [math.NT].
4. **Y. Yuan**, *Descent of (φ, τ)-modules in characteristic p*, 2025. arXiv: 2511.17029 [math.NT].
5. S. Wang and **Y. Yuan**, *On the p-adic transcendence of $\sum_{k=1}^{\infty} p^{-1/p^k}$*, 2025. arXiv: 2509.24609 [math.NT].
6. **Y. Yuan**, *Formalization of Harder-Narasimhan theory*, 2025. arXiv: 2509.19632 [math.AG].
7. **Y. Yuan**, *Multivariable period rings of p-adic false tate curve extension*, 2025. arXiv: 2505.24064 [math.NT].

Books & Textbooks

1. H. Chen, B. Dong, J. Jiang, S. Wang, L. Xiao, **Y. Yuan**, and P. Yu, *Formalized Mathematics & Proof Assistant Lean*. 2026, Textbook in preparation.

Talks

2026, Sept. 18th — **Formalization of Non-Archimedean Functional Analysis**
School of Mathematical Sciences, Fudan University

2026, Jul. 20th — **p-adic mathematics in Mathlib of Lean 4**
Lean 4 Formalization Code Review Training Workshop
Peking University

2026, Apr. 17th — **On the algebraicity of p-adic Hahn series**
Jinling Conference on Number Theory and Arithmetic Geometry
Nanjing University

2026, Apr. 16th — **Formalization of Non-Archimedean Functional Analysis**
Workshop on Formalized Mathematics
Yau Mathematical Sciences Center, Tsinghua University

2026, Apr. 9th — **Formalization of Non-Archimedean Functional Analysis**
School of Mathematical Sciences, East China Normal University

2026, Apr. 7th — **Formalization of Non-Archimedean Functional Analysis**
AI for Mathematics @ SIMIS
Shanghai Institute for Mathematics and Interdisciplinary Sciences

2025, Nov. 19th — **Formalization of Harder-Narasimhan Theorey**
Shanghai Institute for Mathematics and Interdisciplinary Sciences

2025, Jun. 13th — **Arithmetic of spherically complete valued fields in mixed characteristic**
Symposium on Number Theory and Algebraic Geometry of Spring 2025
Capital Normal University

Awards

2024 Outstanding Teaching Assistant in Undergraduate Fundamental Courses, Tsinghua University (Second Class)

2023 Scholarship of Yau Mathematical Sciences Center (Second Class)

Tsinghua University Comprehensive Excellence Scholarship (Second Class)

Peer Review

2024 Journal of Number Theory

Teaching Experience

2026, Autumn TA for **Foundation of Algebra and Analysis (I)** at Westlake University. AI4Math training for undergraduates.

2026, Spring TA for **Foundation of Algebra and Analysis (II)** at Westlake University. Formalizatation (Lean 4) training for undergraduates.

2025, Autumn TA for **Foundation of Algebra and Analysis (I)** at Westlake University. Formalizatation (Lean 4) training for undergraduates.

2024, Autumn TA for an undergraduate course on **Linear Algebra** at Tsinghua University.

2023, Autumn TA for an undergraduate course on **Linear Algebra** at Tsinghua University.

2022, Autumn TA for an undergraduate course on **Linear Algebra** at Tsinghua University.

Skills

Coding Lean 4, Python (SageMath), LaTeX, Mathematica, …